2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 6

Due Date: Due on 5/Nov/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student number**. Write neatly on **A4-sized** paper (*staple if necessary*) and **show your steps**.
- **Late submissions or answers without steps won't be graded.**

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1. Find the value of the determinant of the following matrices.

(1) $\begin{pmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{pmatrix}$ (2) $\begin{pmatrix} x+a_1 & a_2 & \cdots & a_n \\ a_1 & x+a_2 & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ a_1 & a_2 & \cdots & x+a_n \end{pmatrix}$.

2. Find the inverse matrix of the rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

3. Let A and B be $n \times n$ matrices. Prove that if $AB = I$, then $BA = I$.
What is the significance of the result in terms of the definition of a nonsingular matrix?

4. If A is nonsingular, find out the product $A\text{adj}(A)$. What if A is singular?

5. (Bonus!) Let each $a_{ij}(t)$ be a differentiable function of t. Show that

$$\frac{d}{dt}\begin{vmatrix} a_{11}(t) & \cdots & a_{1j}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & \cdots & a_{2j}(t) & \cdots & a_{2n}(t) \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1}(t) & \cdots & a_{nj}(t) & \cdots & a_{nn}(t) \end{vmatrix}$$
$$= \sum_{j=1}^{n} \begin{vmatrix} a_{11}(t) & \cdots & \frac{\mathrm{d}}{\mathrm{d}t}a_{1j}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & \cdots & \frac{\mathrm{d}}{\mathrm{d}t}a_{2j}(t) & \cdots & a_{2n}(t) \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1}(t) & \cdots & \frac{\mathrm{d}}{\mathrm{d}t}a_{nj}(t) & \cdots & a_{nn}(t) \end{vmatrix}$$

and evaluate $\frac{d}{dt}|I + tA|$ when $t = 0$.

6. (a) Use Cramer's rule to solve the following systems.

(b) Find A^{-1} using the adjoint of A. Then solve the system by computing $\mathbf{x} = A^{-1}\mathbf{b}$.

$$\begin{cases} x_1 & +2x_2 & = 3 \\ 3x_1 & -x_2 & = 1 \end{cases}, \qquad \begin{cases} x_1 & +2x_2 & +x_3 = 1 \\ & -x_2 & +x_3 = 2 \\ 2x_1 & +3x_2 & -2x_3 = 3 \end{cases}.$$

7. Let C be the set of complex numbers. Define addition on C by

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

and define scalar multiplication by

$$\alpha(a + bi) = \alpha a + \alpha bi$$

For all real numbers α. Show that C is a vector space over $\mathbb{R}$ with these operations.

8. Let V denote the set

$$\{p(x) |\ p(0) = 1,\ p(x) \in P\}$$

where P denotes the set of all polynomials of any finite degree with real number coefficients. Under the standard addition and scalar multiplication for functions, determine whether V over $\mathbb{R}$ is a vector space.